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### Systems and methods for collision avoidance for shape changing machines

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#### Abstract

A collision avoidance system and method for a mobile machine includes determining a boundary zone and providing a notification and/or control command to prevent a collision with another object. A first safety zone associated with a first and second portion of the mobile machine is determined. A second safety zone associated with the second portion of the mobile machine is determined. A boundary zone is determined as a function of a combination of the first and second safety zone. A notification and/or a control command is provided based on a relationship of an object to the boundary zone.

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## Background/Summary

### TECHNICAL FIELD

(1) The present disclosure relates generally to systems and methods for collision avoidance for shape changing machines, and more particularly, to systems and methods for determining an object's proximity to a determined safety boundary.

### BACKGROUND

(2) Many activities require or benefit from collision detection and avoidance systems. For example, operators of large shape shifting machinery must navigate job and construction sites, not only while driving the machine, but also while operating extensions of the machine. For example, the operator of an excavator machine must be mindful of building structures, materials, people, or other large vehicles when driving and operating the bucket, arm, and boom of the excavator. Failure to do so may result in collision related accidents that can cause extensive and serious damage including property damage causing delays, or serious personal injury. In order to reduce the likelihood of accidents or collisions, operators of large machinery often rely on collision detection and avoidance systems.

(3) Typically, Collision Alarming and Avoidance Systems (CAAS) fall into two categories. The first type of CAAS relates to large machinery that does not change shape. Using simple calculations, this type of system defines "zones" based on the machines' boundaries, safety tolerances, and stop distances. While this type of system is ideal as it requires limited computational power allowing for easy installment on machinery, this type of system is inadequate for shape shifting machines, such as excavators and shovels, as they fail to consider rotational and shape changes when defining a safety zone.

(4) The second type of CAAS applies complex derivative functions and considers multiple variables (vehicle/object dynamics, speeds, steering, etc.) to calculate multiple trajectories for multiple points of the machine. Using this information, the system must then calculate the probabilities of each possible collision and then filter out potential collisions that do not meet a probability threshold. This second type of system requires high-level computational power and sensors which may not be compatible with small computers and engine control modules (ECM) common on some shape shifting vehicles.

(5) Korean Patent Application Publication No. KR 2021 0047 985, published on May 3, 2021 ("the '985 publication"), describes a collision avoidance system for construction equipment. The system

disclosed in the '985 publication uses information of a detected moving obstacle, swing dynamics, and driving operation information, to calculate a collision time with the obstacle. The '985 publication uses a series of equations and derivatives and is an example of the type of CAAS that requires high level computational power and resources.

(6) The disclosed method and system of the present disclosure may solve one or more of the problems set forth above and/or other problems in the art. The scope of the current disclosure, however, is defined by the attached claims, and not by the ability to solve any specific problem.

## SUMMARY

(7) In one aspect, a collision avoidance system for a mobile machine, comprises determining a first safety zone associated with a first and second portion of the mobile machine and a second portion of the mobile machine that is movable relative to the first portion of the mobile machine, determining a second safety zone associated with the second portion of the mobile machine, determining a boundary zone as a function of a combination of the first and second safety zone, and providing a notification and/or a control command based on a relationship of an object to the boundary zone.

(8) In another aspect, the system may include that the first and second safety zones are geometric shapes. In another aspect, the geometric shapes may include one of a rectangle or sector.

(9) In one aspect, the mobile machine is an excavator and the first portion is a propelled chassis of the machine, and the second portion is an arm assembly of the mobile machine. In another aspect, the first safety zone is a function of a speed and movement direction of the first and second portion of the machine. Alternatively, another aspect may include that the second safety zone is a function of the speed and movement direction of the second portion of the machine.

(10) In one aspect, the first safety zone is a function of outputs from a first position sensor associated with the first portion of the mobile machine and a second position sensor associated with the second portion of the machine, and the second safety zone is a function of outputs from the second position sensor associated with the second portion of the machine. In another aspect, the first and second safety zones are also a function of GPS position data. Additionally, in another aspect, providing a notification or control command includes an emergency stop control command.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

(2) FIG. 1 is a diagram illustrating an exemplary environment of a vehicle system for collision avoidance, according to aspects of the disclosure.

(3) FIG. 2 illustrates a diagram of an exemplary controller system of the vehicle system of FIG. 1.

(4) FIG. 3 provides a flowchart depicting an exemplary method for the vehicle system of FIG. 1.

(5) FIG. 4 illustrates a diagram of an exemplary embodiment of the vehicle system of FIG. 1, the controller system of FIG. 2, and the method of FIG. 3.

### DETAILED DESCRIPTION

(6) Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed. As used herein, the terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. In this disclosure, unless stated otherwise, relative terms, such as, for example, “about,” “substantially,”

and “approximately” are used to indicate a possible variation of  $\pm 10\%$  in the stated value.

(7) FIG. 1 is a diagram of illustrating an exemplary environment of an excavator **100**, according to aspects of the disclosure. While this disclosure will be described in association with an excavator **100**, other mobile machines are contemplated, such as, for example, articulated/dump trucks, backhoe loaders, mining shovels, cold planers, dozers, skid steer loaders, motor graders, etc.

(8) The excavator **100** may include a lower portion or lower frame **111**, an upper portion or upper frame **113**, and an arm assembly **115** that all can move relative to one another. The lower frame **111** may include an undercarriage including, for example, a pair of tracks **112** and a track frame **109**. The upper frame **113** may include, for example, a cab **117** and engine **119**. The upper frame **113** may be rotationally coupled to the lower frame **111** so as to rotate independent of tracks **112**. For example, although the travel direction of excavator **100** may be north, upper frame **113** may be facing northwest. The relative rotational movement between upper frame **113** and lower frame **111** is an example of the shape-shifting ability of excavator **100**.

(9) Arm assembly **115** may be coupled to the upper frame **113** to controllably extend towards and away from the upper frame **113**. The arm assembly **115** of excavator **100** may include a bucket **102** used for digging and pushing ground materials, or for holding and transporting materials. Materials, such as dirt or gravel, may be transported in bucket **102** or, alternatively, heavy materials such as concrete piping may be strapped to bucket **102** and hoisted/transported as such. The arm assembly **115** of excavator **100** may also include a stick **104** and a boom **106**, both of which may provide hydraulic support and mobility for bucket **102**. Extension and retraction of the arm assembly **115** may be accomplished by maneuvering the position of stick **104** and boom **106**. The extension and retraction of the arm is another shape-shifting ability of excavator **100**.

(10) Excavator **100** may also include a variety of sensors, controllers, and/or transceivers/receivers that provide readings, measurements, and/or data related to excavator **100**. In an exemplary embodiment, a position sensor may include a bucket inertial measurement unit (IMU) **120** that determines a curl of a bucket or implement positioning. A camera **122** may be provide data for identifying objects, such as object **116**, such data including video or image monitoring. A global positioning system, or GPS **124**, may utilize signals in conjunction with satellite locations to provide accurate positioning, timing, and navigational information regarding excavator **100**. Another positioning sensor may be a chassis IMU **126** that determines the angular velocity or acceleration of chassis **110** as well as the angular velocity or acceleration of arm assembly **115**. Other sensors **128** represents a variety of sensors that may be commonly utilized to determine information useful for position, navigation and/or object detection. Examples of other sensors **128** may include boom and stick IMUs, LIDAR, radar, infrared, stereo/tri lens cameras, etc.

(11) As will be discussed in greater detail in FIG. 2, a controller **220** may receive input information from the various sensors and transceivers/receivers, and process the information using processors and/or memory devices to determine various outputs related to excavator **100**.

(12) In an exemplary embodiment, excavator **100** may be operated by an operator **108**. However, the excavator could also be autonomous or semi-autonomous. In an exemplary embodiment, operator **108** may utilize a display **114** that provides a graphical user interface (GUI) **140**. In this exemplary embodiment, the graphical user interface (GUI) **140** is depicted as a map relative to excavator **100**. The graphical user interface (GUI) **140** may include representations relative to a GUI excavator **148** such as a GUI object **146** and a GUI boundary zone **142**. In this exemplary embodiment, the GUI object **146** is a representation of a pickup truck indicated as object **116**. Although object **116** is moving in this exemplary embodiment, object **116** may be stationary and may include objects of various sizes and shapes. Furthermore, in this exemplary embodiment, GUI boundary zone **142** is a representation of boundary zone **118**. Boundary zone **118** may be a zoned area depicting an area where an object collision may occur. Boundary zone **118** may be determined in a variety of ways and may result in a variety of geometric shapes as will be discussed in greater detail below.

(13) Alert **130** may be used to provide operator **108** with a variety of notifications. The types of notifications may relate generally to object detection or collision avoidance and will be discussed in greater detail below. The alert may be visual, such as by a flashing light indication, audible, or a combination of both.

(14) FIG. 2 illustrates a diagram of a controller system **200**. The controller **220** receives inputs **202**, processes the input information, and provides one or more outputs **228**. Controller **220** may embody a single microprocessor (CPU) or multiple microprocessors, a memory **222**, a secondary storage device, and/or any other means for accomplishing a task consistent with the present disclosure. The memory **222** or secondary storage device associated with controller **220** may store data and/or software routines that may assist controller **220** in performing its functions, such as the functions of method **300** of FIG. 3. Further, the memory **222** or secondary storage device associated with controller **220** may also store data received from the various inputs **202** associated with controller system **200**. Numerous commercially available microprocessors can be configured to perform the functions of controller **220**. Controller **220** may embody a non-transitory machine-readable medium that stores information that, when executed by controller **220**, causes the controller system **200** to perform instructions and may be computer-implemented. It should be appreciated that controller **220** could readily embody a general machine controller capable of controlling numerous other machine functions. Alternatively, a special-purpose machine controller could be provided. Further, one or more operations of controller **220**, may be performed remote from excavator **100** and communicated to excavator **100**. Various other known circuits may be associated with controller **220**, including signal-conditioning circuitry, communication circuitry, actuation circuitry, and other appropriate circuitry.

(15) Inputs **202** to controller **220** may include, for example, chassis speed data **204** (traveling speed and direction of upper frame **113** and lower frame **111**) received from chassis IMU **126**. Using the other information in combination with the speed and direction determined from the chassis IMU **126**, other information may be obtained such as attitude, orientation, position, velocity, acceleration and rotation speed of the chassis of excavator **100**. Similarly, bucket speed data **206** may be received in controller **220** using the angular velocity or acceleration determined from the chassis IMU **126**. Other bucket information such as attitude, orientation, position, velocity, acceleration and rotation speed may also be obtained. Location data **208** may be determined using information obtained from GPS **124**. Optionally, location data **208** may be received by using a navigational map system and initially indicating a start point, either manually, or using GPS **124**. Object detection data **210** may be determined using, for example, camera **122**, LIDAR, radar, infrared, or other sensory/information gatherers. As an exemplary embodiment, camera **122** may use motion detection to identify moving objects. Or alternatively, a radar system may be utilized to identify stationary objects based on sent and reflected energy waves. Other means of detecting objects may be included while the machine is in motion or stopped, as understood and appreciated in the art.

(16) Operator input data **212** may be manually entered by operator **108** and may relate to a variety of information. For example, an exemplary embodiment of operator input data **212** may include operator preferences when determining boundary zone **118**. For example, the operator may prefer that boundary zone **118** is defined by the safety zone of the upper frame **113** and lower frame **111** plus the swing zone of the arm assembly **115**. Alternatively, operator preferences may include a boundary zone **118** that is defined by a rectangle that encompasses the safety zone of the upper frame **113** and lower frame **111** and swing zone of the arm assembly **115**. In another exemplary embodiment, operational preferences may be related to alert **130**. Operator **108** may, for example, be required to provide operational preferences as to whether alert **130** is to be visual, such as by a flashing light indication, audible, or a combination of both audio and visual. Operator input data **212** may also relate to desired safety distances. Excavator **100** may come with a preset safety distance of 20 feet. Alternatively, operator **108** may input the safety distance to be 30 feet. Operator input data **212** may include input from operator **108** designating which devices may be utilized for

navigation. For example, operator **108** may provide operator input data **212** designating that GPS **124** and camera **122** are to be utilized to determine navigation and orientation. Alternatively, operator input data **212** may include a default setting that designates a radar system is to be utilized to identify stationary objects. Operator input data **212** may include other operational preferences that may override default operational preferences with an input by operator **108**.

(17) Inputs **202** to controller **220** may also include braking response time delay data **214**. In one example embodiment, braking response time delay data **214** may relate to the distance and time a machine will travel in relation to the detection of an object. For example, braking response time delay data may include a point in time when an initial object detection is made by a perception sensor such as a camera **122**, LIDAR, radar, infrared, or other sensory/information gatherers. Braking response time delay data **214** may also include time elapsed and a distance travelled from when the device first detects the object and time taken to confirm the object exists. Braking response time delay data **214** may also include time elapsed and a distance travelled while determining a risk level and selecting and transmitting a course of action. When the course of action is selected, braking response time delay data **214** may also include time elapsed and a distance travelled to account for the reaction time of operator **108** or controller **220** engaging the braking system. Braking response time delay data **214** may also include time elapsed and a distance travelled while braking. When the machine comes to a final stopping point, braking response time delay data **214** may also include a boundary zone **118**, as will be discussed in greater detail below.

(18) Controller **220** may utilize inputs **202** and combine it with information in memory **222**. Memory information may include a variety of information that may be associated with excavator **100**. For example, memory information may include the geometry of the machine or information related to the machine's dimensions. In one exemplary embodiment, the information of the geometry of excavator **100** may include track dimensions, implement dimensions, chassis dimensions, arm dimensions and geometry (e.g., maximum length of the arm, rotational arc of the arm based on the position of boom **106** and stick **104**), or other relevant information.

(19) Information in memory **222** may also include a series of maps or data charts. In one example embodiment, a map may include an emergency stop map that correlates the speed or acceleration of excavator **100** to a stopping distance. For example, an emergency stop map may include speeds ranging from 1 to 25 mph and a corresponding distance that excavator **100** will continue to travel before coming to a complete rest for each specific speed value. The emergency stop map may include that at 5 mph, the distance traveled by excavator **100** before coming to a complete stop will be 10 feet. The emergency stop map may also provide that at 10 mph, the distance traveled by excavator **100** before stopping will be 15 feet.

(20) The emergency stop map may be one dimensional, such as in the example embodiment recently described, or it may be two dimensional and include another variable to correlate with a stopping distance. For example, another exemplary embodiment may include a two dimensional emergency stop map that correlates the speed or acceleration of excavator **100**, along and a load amount in the bucket, to a stopping distance. As an example, a two dimensional emergency stop map may include that at 5 mph and with 1500 pounds in the bucket, the distance traveled by excavator **100** before stopping will be 20 feet. Or a two dimensional emergency stop map may include that at 10 mph and with 500 pounds in the bucket, the distance traveled by the machine before stopping will be 17 feet. Other variables known in the art may be used to create multiple one or two dimensional emergency stop maps.

(21) Information in memory **222** may also include an arm assembly stop map that correlates the rotational speed or rotational acceleration of the upper frame **113** and arm assembly **115**, to a stopping rotational distance. For example, an arm assembly stop map may include speeds ranging from 1 to 10 mph and a corresponding distance the arm assembly **115** will continue to rotate, in degrees or radians, before coming to a complete rest for each specific rotational speed/acceleration value. The arm assembly stop map may include that at a rotational speed of 5 degrees per second,

the distance traveled by the bucket and assembly arm, before coming to a complete stop, will be 20 degrees. The arm assembly stop map may also include that at 10 degrees per second, the distance traveled by bucket and assembly arm before coming to a complete stop, will be 27 degrees. Other rotational distances and corresponding rotational speeds may also be included in an arm assembly stop map.

(22) Alternatively, in another exemplary embodiment, an arm assembly stop map may be a two dimensional map and include another variable to correlate with the rotational stopping distance. In this example embodiment, a two dimensional arm assembly stop map may utilize the rotational speed or acceleration of the upper frame **113** and arm assembly **115**, along with a load amount in the bucket, to provide a rotational stopping distance of the bucket and arm assembly. As an example, a two dimensional arm assembly stop map may determine that the arm assembly/bucket will rotate 30 degrees if it has a rotational speed of 10 degrees per second and 1000 pounds of materials in the bucket. Alternatively, a two dimensional arm assembly stop map may include that the arm assembly/bucket will rotate 40 degrees if it has a rotational speed of 10 degrees per second and 2000 pounds of materials in the bucket. Other rotational distances corresponding to rotational speed and bucket load may also be considered in a two dimensional arm assembly stop map. Furthermore, other variables known in the art may be considered in a one or two dimensional arm assembly stop map.

(23) In addition, memory **222** may include safety margins. Safety margins may be related to the travel direction of excavator **100** and provide an extra added safety distance when determining possible collisions. For example, memory **222** may require that a safety margin of 5 feet in all directions be included. Alternatively, a safety margin may be related to the rotation of the arm assembly **115** and bucket **102**. A safety margin related to arm assembly **115** and bucket **102** may require that a safety margin of an additional 30 degree rotation be factored into an emergency stop zone for avoiding possible collisions. Alternatively, safety margins may be expressed in terms of time and require that a machine stop within 1 second of collision. As time is a function of distance and rate, the safety margins are similar and require little calculations to convert from one expression to the other.

(24) Processing the inputs **202** in controller **220** may allow controller **220** to determine the outputs **228**, including, for example, a boundary zone display **230**, a control command **232**, and/or notification **234**. Controller **220** may provide additional outputs in addition to those provided herein. Boundary zone display **230** may represent boundary zone **118** and be displayed to an operator **108** as GUI boundary zone **142** on a GUI display **114** (see FIG. 1). Determining boundary zone **118** will be discussed in greater detail below in connection with FIG. 3.

(25) In another exemplary embodiment, a control command **232** may be one of outputs **228**. In another exemplary embodiment, control command **232** may include commands that are related to collision avoidance when it is detected that an object traverses boundary zone **118**. For example, a control command may include an emergency stop that automatically terminates operation of excavator **100** by killing all power to any power trains, hydraulic systems, or engines and removing operational control from operator **108** when controller **220** determines that an object traverses boundary zone **118**. Alternatively, another exemplary embodiment of a control command **232** may include disabling most operational controls from operator **108** and placing the machine in a standby mode when it is determined that an object is to traverse boundary zone **118**. Or in another example embodiment, a control command **232** may include removing or suspending all power to excavator **100** until operator **108** provides an override indication in order to continue to operate the machine. In another example embodiment, outputs **228** may include a control command **232** that overrides steering controls by operator **108** and steers excavator **100** to avoid collision with another object. For example, outputs **228** may include a control command **232** to steer excavator **100** left to avoid colliding with an object in front of excavator **100**. In another exemplary embodiment, a control command **232** may include overriding controls made by operator **108** related to the arm of



excavator **100**. For example, a control command **232** may be to restrict movement of the arm of excavator **100** despite the controls of operator **108** if restricting movement of the arm assembly **115** avoids collision with an object. Other variations of control command **232** may also be utilized as is common and appreciated in the art.

(26) Outputs **228** of controller system **200** may also or alternatively include a notification **234**. In one exemplary embodiment, notification **234** may be an alert that notifies operator **108** of a potential collision by utilizing alert **130** to provide operator **108** with a visual alert notification, such as a flashing light, or an audible notification, or a combination of both audio and visual. Alternatively, notification **234** may include providing a visual notification to graphical user interface (GUI) **140** of display **114**, such as a flashing or blinking display. Other variations of notification **234** may be utilized as is common and appreciated in the art.

(27) FIG. 3 provides a flowchart of method **300** depicting an exemplary method for collision detection and avoidance. In an exemplary embodiment, the process may include determining a first safety zone associated with a first portion of the mobile machine **302**. The first portion of the machine may include lower frame **111** and upper frame **113**. In this example embodiment, the upper frame **113** includes arm assembly **115** being fully extended, however, other embodiments may consider various levels of extension, related to arm assembly **115**, in determining the first safety zone.

(28) Utilizing information from inputs **202** such as speed and direction of excavator **100**, as well as information in memory **222** such as the emergency stop map and a safety margin, controller **220** may provide a first safety zone. In this example embodiment, the first safety zone may provide a boundary or indication of how far the upper frame **113** and lower frame **111** of excavator **100** will continue to travel in the event that power to excavator **100** is instantaneously removed, plus a safety margin from memory **222**. The safety margin allows for an additional safety distance beyond where the vehicle is predicted to come to a stop. In the instance where upper frame **113** and lower frame **111** are at rest, the speed of excavator **100** would be 0 miles per hour, but the first safety zone would extend beyond the outline of excavator **100** because in this exemplary embodiment, a safety margin is factored into determining the first safety zone.

(29) In this exemplary embodiment, the method **300** may include determining a second safety zone associated with a second portion of the mobile machine, the second portion of the machine movable relative to the first portion of the machine **304**. In this exemplary embodiment, the second portion of the mobile machine may account for the rotation of upper frame **113**, and particularly arm assembly **115**. The second safety zone associated with a second portion of the mobile machine may include an emergency stop zone for arm assembly **115** plus a safety margin. By utilizing inputs **202** relating to the speed and rotational direction of arm assembly **115**, the emergency stop zone for arm assembly **115** may indicate a boundary that identifies the size of a sector (arc of a circle) corresponding to how far the arm assembly **115** of excavator **100** will continue to traverse after an emergency stop (terminating all power to the vehicle) is activated. In this exemplary embodiment, determining the second safety zone may include adding a safety margin from controller **220** and memory **222** that would provide an additional safety distance and buffer beyond where the arm assembly **115** is predicted to stop.

(30) In this exemplary embodiment, the process may include determining a boundary zone as a function of the combination of the first and second safety zone **306**. In this exemplary embodiment, determining boundary zone **118** may include identifying the greater boundary of the first safety zone and the second safety zone combined. This may result in an irregular and unsymmetrical shape, e.g., a rectangular shape corresponding to the first safety zone and a sector corresponding to the second safety zone. This embodiment may result in a boundary zone that follows the perimeter of excavator **100** closely and results in fewer control commands and/or notifications and may be displayed on display **114** to be viewed by operator **108**.

(31) Alternatively, an exemplary embodiment of the boundary zone may be defined by a rectangle

that encompasses both the first and second safety zone. This exemplary embodiment may result in a larger boundary zone and may result in more control commands and/or notifications. Other shapes that encompass both the first and the second safety zone may be considered, such as a circle or square. Additionally, the shape of the boundary zone may be a two-dimensional or three-dimensional polygon. In other exemplary embodiments, the shape of the boundary zone may be any other polygonal, geometrical, symmetrical or irregular, shape.

(32) In this exemplary embodiment, the method may include providing a notification or a control command based on a relationship of an object to the boundary zone **308**. In one example embodiment, the relationship of an object to the boundary zone may include the object crossing or traversing into the boundary zone. Detecting that an object has traversed a boundary zone **118** may be determined visually by operator **108**. Alternatively, the method may utilize sensors or devices to determine an object's speed and travel direction in relation to the boundary zone while the machine is fully stopped. In one exemplary embodiment, the machine may utilize LIDAR by transmitting light waves and measuring the reflected light waves to determine an object's speed and direction. Alternatively, an infrared system may be utilized to detect heat energy from an individual that is walking on the construction site. Other exemplary embodiments for determining an object's speed and direction while the machine is fully stopped, and in relation to the boundary zone, may utilize cameras, radar, or other devices.

(33) In an exemplary embodiment, providing a notification or control command may take place the moment the first object intersects the boundary zone and may include a variety of notification outputs. In one example embodiment, a notification may utilize alert **130** to produce a visible notification, an audible notification, or a combination visible/audible notification as discussed in relation to **234**. In another embodiment, a notification **234** may be provided through the graphical user interface (GUI) **140** of display **114**. The notification **234** may include displaying the boundary zone **118** as a function of the first and second safety zones on the graphical user interface (GUI) **140** of display **114**. An exemplary embodiment may also include display **114** flashing, blinking, or providing textual descriptions relating to the first object entering the boundary zone **118**. In another embodiment, a color notification **234** may be provided on display **114** or by utilizing alert **130**. For example, a notification **234** may include a yellow warning flash indicating that a collision may occur. Alternatively, a notification **234** may include a red warning flash indicating that a collision is imminent. In another exemplary embodiment, a notification **234** may be a grey flash indicating that another object is present and may potentially be a collision threat. Other types of notification **234** known in the art may also be included with other exemplary embodiments.

(34) Providing a control command **232** may include a variety of instructions from the controller **220**. As was previously mentioned, one example embodiment may include providing a control command **232** that outputs an emergency response causing the system power of excavator **100** to abruptly cease. Other control commands may exist, including those previously discussed in relation to control command **232** and those that are known in the art, which may be also considered in additional embodiments.

#### INDUSTRIAL APPLICABILITY

(35) The disclosed method for collision detection and avoidance may be used on any suitable machinery capable of colliding with other objects. An operator **108** may use a collision detection and avoidance system to ensure safety when operating mobile machinery. Controller system **200** may process, in real-time, data and informational inputs **202** to help prevent collisions by providing a boundary zone display **230**, executing a control command **232**, and/or providing a notification **234**. The disclosed method for collision detection and avoidance may find particular applicability in machines operating in industries including construction, mining, farming, etc. Non-limiting examples of machines may include commercial machines, such as excavators and other earth moving vehicles.

(36) FIG. 4 illustrates an exemplary embodiment of the operation of controller system **200** and the

flow chart of FIG. 3. Diagram 400 depicts boundary zone 118 at two different points in time, T=0 and T=1. In this embodiment, controller 220 may have information from inputs 202 and information in memory 222. Inputs 202 may include chassis speed data 204 (traveling speed and direction of upper frame 113 and lower frame 111) received from chassis IMU 126. In this example, the controller 220 is able to process the data inputs 202 and determine that excavator 100 is traveling at a speed of 3 miles per hour (mph) in a direction from east to west. Other inputs 202 may include bucket speed data 206 received from chassis IMU 126 and may be used by controller 220 to determine an angular velocity and or other information such as attitude, orientation, position, velocity, acceleration and rotation speed. In this example, using chassis IMU 126, controller 220 may determine that arm assembly 115 is rotating at a rotational speed of degrees per second clockwise.

(37) Controller 220 may also determine a first and second safety zone, and a boundary zone by combining the first and second safety zones. With respect to the first safety zone corresponding to the linear travel distance of the arm assembly 115, excavator upper frame 113, and lower frame 111 (not pictured), controller 220 may utilize informational inputs 202 and memory 222 to determine a stop zone 406 indicating the path that excavator 100 will travel if an emergency stop control command 232 were triggered. In this exemplary embodiment, controller 220 may utilize an emergency stop map to determine that excavator 100 will travel 20 feet before stopping at the boundary of stop zone 406. Controller 220 may utilize memory 222 to determine a first safety zone 408 by including a safety margin added to stop zone 406. In this exemplary embodiment, first safety zone 408 outlines an additional distance from stop zone 406 to provide the additional safety distance. For example, controller 220 may determine the first safety zone 408 by utilizing location data 208 and chassis speed data 204 to determine the speed and/or acceleration of excavator 100. Controller 220 may access memory 222 to add a safety margin of an additional 5 feet from stop zone 406 to determine the boundary for the first safety zone 408. Thus, in this exemplary embodiment, first safety zone 408 would be 25 feet in front of excavator 100 and 5 feet of distance would be added to the back and sides of stop zone 406 to create first safety zone 408.

(38) At T=0, arm assembly 115 is rotating in a clockwise direction at a rotational speed. Input data such as bucket speed data 206, location data 208, and chassis speed data 204 may be utilized to determine the rotational speed and/or acceleration of arm assembly 115 and excavator upper frame 113. An arm assembly stop map may be used to determine a swing zone or second safety zone 410. In this embodiment, second safety zone 410 shows the rotational path as a sector that the arm assembly 115 will traverse in the event that an emergency stop were triggered. For example, an emergency stop while arm assembly 115 is rotating clockwise may result in the arm rotating an additional 90 degrees (approximation). In another exemplary embodiment, a safety margin may also be considered to determine the second safety zone.

(39) At T=0, controller 220 may combine the first safety zone and the second safety zone to form boundary zone 118. In this exemplary embodiment, boundary zone 118 is defined as a rectangle that encompasses first safety zone 408 and second safety zone 410. However, as explained previously, other embodiments may determine boundary zone 118 using alternative methods that produce geometrically different or irregular shapes and/or polygons.

(40) For example, another embodiment's boundary zone may simply be an irregular shape that is the combination of first safety zone 408 and second safety zone 410. In this alternative embodiment, stationary object 422 would not trigger a control command 232 and/or notification 234 at T=0 because stationary object 422 would be outside the boundary zone that is an irregular shape comprising the first safety zone 408 and second safety zone 410.

(41) Referring back to the embodiment of FIG. 4, at T=0, controller 220 may utilize object detection data 210 from camera 122, LIDAR, radar, infrared, or other sensory/information gatherers to determine an object's initial point and traveling direction while the machine is stopped. In this exemplary embodiment, stationary object 422 would have been detected and controller 220

would utilize the object detection data **210**, including the object's initial point and traveling direction, to monitor the situation. In one exemplary embodiment, controller **220** may be prepared to output a control command **232** of an emergency safety stop the moment the detected object were to intersect boundary zone **118**. At the moment of an object intersecting boundary zone **118**, an emergency stop control command **232** would cause all operational movement to cease. Alternatively, or in addition, controller **220** may output a notification **234** such as an audio, visual, or combination audio and visual, to operator **108**.

(42) In the exemplary embodiment at  $T=1$ , arm assembly **115** has rotated approximately degrees and is continuing to rotate in a clockwise direction at the same rotational speed determined from bucket speed data **206**. Second safety zone **410** represents the rotational path across multiple sectors of arm assembly **115** in the event that an emergency stop were triggered. Compared to second safety zone **410** at  $T=0$ , the second safety zone **410** at  $T=1$  has also rotated 15 degrees. In this exemplary embodiment at  $T=1$ , controller **220** may include a control command **232** such as an emergency stop while arm assembly **115** is rotating clockwise because controller **220** may determine using an arm assembly stop map that an emergency stop control command would result in arm assembly **115** rotating an additional 90 degrees (approximation) and striking moving object **420**. Again, in this exemplary embodiment, boundary zone **118** is a rectangle that encompasses first safety zone **408** and second safety zone **410**. In this exemplary embodiment, moving object **420** would trigger controller **220** to perform an emergency stop control command **232** as soon as moving object **420** intersected boundary zone **118**. At the moment of intersecting boundary zone **118**, an emergency stop control command would cause all operational movement to cease. In another embodiment, controller **220** may determine the object's speed and position, or an initial and final location, in relation to boundary zone **118** and perform an emergency stop control command **232** by comparing the object's final location to the boundary zone **118**. Alternatively, as another embodiment may include safety zones in relation to time, the final location of object **420** may be compared to the later time between the first safety zone and the second safety zone when determining when to issue an emergency stop control command **232**. Other embodiments of when to issue a control command **232** may exist and/or include a notification **234** of an audio, visual, or a combination of audio and visual provided to operator **108** for notifying of a potential collision.

(43) Thus, this method is able to prevent and detect possible collisions with other objects for shape shifting, heavy machinery without requiring extensive calculations or high computational power from devices. Instead, collision avoidance is achieved using minimal calculations and standard sensors that are compatible with both small and large/heavy machinery.

(44) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed system without departing from the scope of the disclosure. Other embodiments of the system will be apparent to those skilled in the art from consideration of the specification and practice of the system disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope of the disclosure being indicated by the following claims and their equivalents.

## Claims

1. A collision avoidance system for a machine, comprising: a safety zone system installed on the machine, the safety zone system: determining a first safety zone associated with and encompassing a first and second portion of the machine, the second portion of the machine movable relative to the first portion of the machine such that at least a component of the second portion of the machine is outside of the first safety zone; determining a second safety zone associated with the second portion of the machine, the second safety zone encompassing the component of the second portion which is outside of the first safety zone, such that the second safety zone extends beyond the first safety zone, wherein the component of the second portion is movable between: first and second

positions in a first area outside of the first safety zone, wherein the first area encompasses an entire rotational path of movement of the second portion in a first direction, and third and fourth positions in a second area outside of the first safety zone, wherein the second area encompasses an entire rotational path of movement of the second portion in a second direction, wherein the second safety zone includes either the first area or the second area depending on whether the second portion is moving towards the first area or the second area; determining a boundary zone as a function of a combination of the first and second safety zones; and providing a notification and/or a control command based on a relationship of an object to the boundary zone.

2. The collision avoidance system of claim 1, wherein the first and second safety zones are geometric shapes, and wherein the geometric shapes include one of a rectangle or sector.

3. The collision avoidance system of claim 1, wherein the machine is an excavator and the first portion is a propelled chassis of the machine, and the second portion is an arm assembly of the machine.

4. The collision avoidance system of claim 1, wherein the first safety zone is determined as a function of a speed and movement direction of at least one of the first portion and second portion of the machine.

5. The collision avoidance system of claim 4, wherein the second safety zone is determined as a function of the speed and movement direction of the second portion of the machine.

6. The collision avoidance system of claim 1, wherein the first safety zone is determined as a function of outputs from a first position sensor associated with the first portion of the machine and a second position sensor associated with the second portion of the machine, and the second safety zone is a function of outputs from the second position sensor associated with the second portion of the machine.

7. The collision avoidance system of claim 1 wherein the first and second safety zones are determined as a function of GPS position data.

8. The collision avoidance system of claim 1, wherein the providing of a notification or control command includes an emergency stop control command.

9. The collision avoidance system of claim 1, wherein the second safety zone includes a perimeter, wherein at least a portion of the perimeter is arcuate.

10. A computer-implemented collision avoidance method for a machine, the method comprising: determining a first safety zone associated with and surrounding a first portion of the machine and second portion of the machine, the second portion of the machine movable relative to the first portion of the machine such that at least a first component of the second portion is movable from a location inside the first safety zone to a location outside of the first safety zone; determining that the first component of the second portion is at moving toward the location outside of the first safety zone; determining a second safety zone associated with the first component of the second portion of the machine which is moving toward the location outside of the first safety zone, such that at least a part of the second safety zone extends outside of the first safety zone, wherein the second safety zone includes an entire rotational path of the first component of the second portion outside of the first safety zone, when the first component moves in a first direction; determining a boundary zone as a function of a combination of the first and second safety zones; and providing at least one of a notification and/or or a control command based on a relationship of an object to the boundary zone.

11. The method of claim 10, wherein the first and second safety zones are geometric shapes, and wherein the geometric shapes include one of a rectangle or sector.

12. The method of claim 10, wherein the machine is an excavator and the first portion is a propelled chassis of the machine, and the second portion is an arm assembly of the machine.

13. The method of claim 10, wherein the first safety zone is determined as a function of a speed and movement direction of at least one of the first portion and second portion of the machine.

14. A non-transitory machine-readable medium storing instructions that, when executed by a

computing system, causes the computing system to perform a collision avoidance method for a machine, the method including: determining a first safety zone associated with a first portion and second portion of the machine such that the first and second portions are within the first safety zone, the second portion of the machine movable relative to the first portion of the machine such that the second portion of the machine moves between locations inside and outside of the first safety zone; determining a second safety zone associated with a component of the second portion, the second safety zone corresponding to an entire range of possible movement of the component of the second portion of the machine outside of the first safety zone, when the component moves in a predetermined direction; determining a boundary zone as a function of a combination of the first and second safety zone; and providing a notification and/or a control command based on a relationship of an object to the boundary zone.

15. The non-transitory machine-readable medium of claim 14, wherein the first and second safety zones are geometric shapes, and wherein the geometric shapes include one of a rectangle or sector.

16. The non-transitory machine-readable medium of claim 14, wherein the machine is an excavator and the first portion is a propelled chassis of the machine, and the second portion is an arm assembly of the machine.

17. The non-transitory machine-readable medium of claim 14, wherein the first safety zone is a function of a speed and movement direction of at least one of the first portion and second portion of the machine.

18. The non-transitory machine-readable medium of claim 14, wherein the second safety zone is determined as a function of a speed and movement direction of the second portion of the machine.

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